Al-Balqa Applied University
Faculty of Engineering Technology

General chemistry lab /60 min
Midterm exam 20 mark

24/11/2021 - Wednesday

Student name:	Instructor:	Section:

{Helpful information: Awt: Ca = 40/ C = 12 / H = 1 / S = 32 / O = 16 / N= 14 / Cl = 35.5 g/mol
760 mmHg = 1 atm = 760 torr}

* Choose the correct answer for the following questions and transfer it to the table below:

Question	1	2	3	4	5	6	7	8	9	10
Answer										

1) The correct statement of the following is:
   A- You can heat all types of liquid in blue flame directly.
   B- Graduated cylinder is the most precises volumetric glass wares.
   C- Acid and Bases are example of corrosive materials.
   D- Analytical balance error is ±0.001 g.

2) This pictogram on the right indicates that the chemical material is :
A- Toxic
B- Radio active
C- Oxidizing
D- Flammable



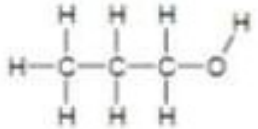
3) Which one of the chemical salts will show precipitate with water?
   A- NH_4^+ Salts
   B- Group 1 carbonates (CO_3)
   C- Silver (Ag) Halides (AgX)
   D- All Nitrates (NO_3)

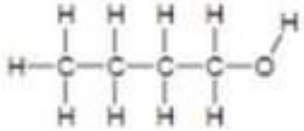
4) During Density experiment, the data in the table beside are obtained, calculate the final volume of water in the cylinder used in this measurement:

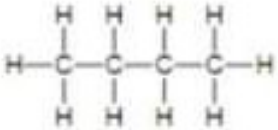
A- 13.78 mL
B- 14.28 mL
C- 17.18 mL
D- 17.68 mL

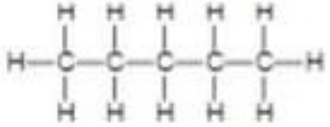
The density of solid	8.62 g/cm^3
Initial volume of water	11.9 ml
Mass of empty cylinder	25.3 g
Mass of cylinder + solid	41.5 g

5) Which one of these compounds give you the highest boiling point:

A- 

B- 

C- 

D- 

6) Which one of the following statements is correct concerning Boiling point experiment:

A- The evolution of air bubbles during boiling point experiment indicates that you reach the equilibrium between atmospheric pressure and vapor pressure of liquid.

B- The normal boiling point of a liquid is the temperature at which the liquid boils at external pressure equals 760 atm.

C- Boiling of liquid takes place at all temperature.

D- If some glass impurities contaminate the liquid, then the boiling point will increase.

7) Calculate the mass of water in $CaSO_4.2H_2O$, if the mass of anhydrous salt = 5.7g.

A- 2.26g B- 1.51g C- 1.99g D- 2.98g

8) A compound is called a hydrated salt when it.....

A- Is composed of only hydrogen and oxygen.

B- Is composed of water and anhydrous salt.

C- Its absorbed water from outside.

D- Its dissolving in water.

9) When 75.06 grams of a compound containing only C, H, and N was decomposed in a combustion furnace, 203.85 grams of CO_2 and 76.35 grams of H_2O were produced according to the chemical reaction below. What is the empirical formula of this compound? $C_xH_yN_z + O_2 \rightarrow CO_2 + H_2O + NO_2$

A- $C_5H_{11}N_2$ B- $C_6H_{11}N$ C- $C_6H_{13}N_2$ D- $C_5H_{13}N_2$

10) A compound has an empirical formula of $C_3H_6Cl_2$. Its molar mass is 226 g/mol. Find its molecular formula.

A- $C_3H_6Cl_2$ B- $C_6H_6Cl_2$ C- $C_6H_{12}Cl_4$ D- $C_6H_{12}Cl_2$

Gd Lu C K